

SYLLABUS FOR PRECLINICAL STUDIES

MODULE I

BODY STRUCTURE, ORGANIZATION AND FUNCTIONS-II (BSOF 1012)

MODIFIED MARCH, 2022

MODULE INFORMATION

- I. Module Title:** Body Structure, Organization and Function-II
- II. Module Code:** BSOF 1012
- III. Credit Value (ECTS):** 6
- IV. Duration:** Year I, Semester I
- V. Prerequisites:** Premedical courses
- VI. Module Description**

This module is designed for first year undergraduate medical students and is intended to describe body structure and function, as well as molecular mechanisms that operate in the living cell. It also deals with developmental periods, with an emphasis on the beginning of human development, formation of embryonic discs and organogenesis, the fetal period, human birth defects, formation of all body systems and molecular regulation of development.

VII. Module Contents and Learning Outcomes

Module objectives:

At the end of the module the preclinical students will be able to:

- Recognize the body's structure, organization and function, including molecular mechanisms that operate in the living cell
- Dissect cadavers, use and operate light microscopes, learn basic histological techniques and perform biochemical tests in the living system
- Recognize that physiology is a science involving adjustment mechanisms to prevailing environmental conditions and understand the concepts of homeostasis, allostasis and heterostasis

Supportive objectives of the Module

At the end of the module the preclinical students will be able to:

- Interpret the potential spaces in the body cavities and the arrangement of their membranes
- Describe the structure and functions of skin, superficial fascia and subcutaneous tissue
- Compare the processes of bone formation and resorption
- Explain the components of the skeletal muscles of the back and upper limb
- Follow the guidelines for dissection of the back and upper limb regions
- Trace the proximal and distal attachments, innervations and vascularization of each muscle in the back and upper limb too

- Trace the vertebrae, curvatures, ligaments and joints of the backbone
- Describe the sites of safe lumbar puncture
- Describe common bone fractures and their management
- Identify the superficial, intermediate and deep back muscles, along with their blood and nerve supply
- Identify the anterior and posterior compartment muscles of the upper limb, along with their blood and nerve supply
- Observe how tissue sections are prepared for light microscopic examination
- Describe the organization of cells and extracellular matrix in the tissues, organs and organ systems
- Demonstrate knowledge of the sequence of development from fertilization, formation of tissue organs and organ systems
- Identify developmental malformations of organs and organ systems.
- Demonstrate understanding of the interaction of cells for induction, determination and differentiation in the formation of organs and organ systems, as well as molecular regulation for such interactions
- Describe genetic and environmental factors that govern the development of the human body
- Describe the basic principles of physiological adjustment.
- Describe the distribution of fluid and electrolyte across cell membranes
- Describe transport mechanisms across cell membranes
- Describe the autonomic nervous system
- Understand the bioelectricity of cells
- Explain autonomic control of visceral function

VIII. Learning and Teaching Strategies

8.1. The strategies include:

- Small group learning
- Practical learning
- Demonstrations

8.2. Methods of Teaching and Learning:

- Standard Lecture
- Seminars
- Tutorials
- Demonstration
- Practical learning
- Dissections

8.3. Human resources:

- **Staff:** Instructors and graduate assistants drawn from the department of anatomy, and physiology will be involved in lecture delivery, laboratory demonstration, facilitation and assessment of seminars, tutorials, quizzes and tests.
- **Spaces/location:** lecture halls (Sefere Selam), labs, dissection rooms (Black Lion)

- **Instructional media:** interactive white board, printed materials (handouts, books, and work sheets), blackboards, display board, flip charts, slides, overheads, real objects or models, images, projector, LCD projectors, DVDs and CD-ROMs will be used during lectures, seminars and tutorials. The laboratory session requires ball and stick models of different atoms, pH meters and solutions of various pH strengths for biochemistry; forceps, scissors, blades, wood block, dissection table etc for anatomy; microscope, images for histology; preserved embryonic specimens for embryology.
- **Time:** Lecture: Monday to Friday from 8:30-12:10 a.m.
Laboratory: Monday to Friday 1:30 - 04:30 p.m.

IX. Role of Instructors:

The role of instructors includes giving standard lectures, seminars and assignment; involving learners fully in assessments, help students to understand what is expected, addressing students questions, facilitating discussion, and developing learners roles in moderation activities. It is also to instructors' role to plan, design and carryout the lesson plan, to collaborate with physiology staff to accomplish the module successfully, to dissect cadavers, demonstrate prosected cadaver anatomy, to evaluate students and to correct quizzes, tests and exams.

X. Role of learners:

It is the learners' role to engage actively in learning, attend lectures, participate actively, share ideas with each other, ask questions on unclear issues, do seminars and assessments, choose topics for tutorials, refer to different reading materials, review their own learning through self-assessment and to demonstrate their knowledge, skill, attitudes and capabilities through a range of evidence including self-assessment, quiz, tests, activities, seminars and examination.

XI. Assessment methods

Assessment criteria of anatomy, embryology, histology, and physiology

- Seminar/assignment/ practical oral exam - 10%
- Two tests – 40%
- Final examination- 50%

NB: The above assessment methods are open for changes.

Body Structure, Organization and Function-II

Module Name	Body Structure, Organization and Functions-II
Module number	BSOF 1012
ECTS	6
Coordinator	Yossef Teshome (Department of Anatomy)
Collaborator	Mrs. Tigst Kena (Department of Physiology)
Course Schedule	March 14- April 22/2022
Duration of the course	6 weeks including exam week
Number of exams	Two progressive tests (50 questions each) One oral exam (Tentative) Final exam (100 questions)
Placement for lectures	Sefere Selam campus
Practical placements	DR (Anatomy department Room-I and Room-II) Histology lab (Anatomy department Room-I)

COURSE CALENDAR FOR MODULE-I: BSOF-II

MODULE I: BSOF II (Week 1-6)

MODULLE ONE: BSOF II THEME ONE TIME TABLE FOR WEEK ONE (March 14- March 18/2022)					
TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
8:30:-10:10 AM INSTRUCTOR	Anatomy: Back anatomy: General description; key features; components. Mr Yossef	Anatomy: Back musculature, and Neurovasculature bundles Mr Yossef	Anatomy: Introduction to the upper limb, Bones of the upper limb and Joints Mr Solomon A	Anatomy: Pectoral region & the Shoulder region, Mr Solomon A	Histology: Epithelial Tissue1: Form & Characteristics Prof. Mekbeb
10:10-10:30 AM	BREAK				
10:30:-12:10 AM INSTRUCTOR	Anatomy: Typical and atypical vertebrae. Back ligaments and joints Mr Yossef	Anatomy: Clinical correlation; Lumbar puncture; disc prolapse Mr Yossef	Anatomy: Superficial structures of upper limb & Breast Mr Solomon A	Anatomy: Axilla, Boundaries, Contents, communication s Mr Solomon A	Histology: Covering & Glandular Epithelium Prof. Mekbeb
12:00-1:30 PM	LUNCH				
1:30-4:30 PM INSTRUCTOR	DR1: Group 1 DR2: Group 2 Mr. Solomon & Mr. Fikre	DR1: Group 3 DR2: Group 4 Mr Assaye & Mr. Yossef	DR1: Group 5 DR2: Group 6 Mr. Fikre & Mr. Assaye	DR1: Group 1 DR2: Group2 Mr. Yossef & Mr. Fikre	DR1: Group 3 DR2: Group 4 Dr. Girma & Mr Solomon
INSTRUCTOR					

MODULLE ONE
BSOF II
TIME TABLE FOR WEEK TWO (March 21 – March 25/2022)

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
8:30:-10:10 AM	Anatomy: Anatomy: Axilla, Boundaries, Contents, communications	Anatomy: Forearm (Anterior & Posterior compartments)	Histology: Connective tissue proper 1	Histology: Connective tissue 2: Special CT	Histology: Nervous Tissue 1: Neurons 2: Neuroglia
INSTRUCTOR	Mr Solomon A	Mr Solomon A	Prof. Mekbeb	Prof. Mekbeb	Prof. Mekbeb
10:10-10:30 AM	BREAK				
10:30:-12:10 AM	Anatomy: Arm (Anterior & Posterior compartments& Cubital fossa	Anatomy: Hand (dorsum and palm of the hand)	Histology: Connective tissue proper 1:	Histology: Connective tissue 2: Special CT	Histology: Nervous Tissue 1: Neurons 2: Neuroglia Peripheral Nervous System
INSTRUCTOR	Mr Solomon A	Mr Solomon A	Prof. Mekbeb	Prof. Mekbeb	Prof. Mekbeb
12:10-1:30 PM	LUNCH				
1:30-4:30 PM	DR1: Group 5 DR2: Group 6	DR1: Group 1 DR2: Group 2	DR1: Group 3 DR2: Group 4	DR1: Group 5 DR2: Group 6	DR1: Group 1 DR2: Group 2
INSTRUCTOR	Dr. Girma & Mr. Abay	Mr. Yossef & Mr Assaye	Mr. Solomon & Mr. Yossef	Dr. Girma & Mr. Fikre	Mr. Assaye & Dr. Soressa
INSTRUCTOR	Hist. Lab 1: Group 1 1:30- 3:30 PM Hist. Lab 2: Group 2 3:30- 5:30 PM Dr. Soressa	Hist. Lab 1: Group 3 1:30- 3:30 PM Hist. Lab 2: Group 4 3:30- 5:30 PM Mr. Fikre	Hist. Lab 1: Group 5 1:30- 3:30 PM Hist. Lab 2: Group 6 3:30- 5:30 PM Mr. Fikre	Hist. Lab 1: Group 1 1:30- 3:30 PM Hist. Lab 2: Group 2 3:30- 5:30PM Mr. Solomon	Hist. Lab 1: Group 3 1:30- 3:30 PM Hist. Lab 2: Group 4 3:30- 5:30 PM Mr. Solomon
1:30-4:30 PM			Group 1 Ortho Group 2 Surgery	Group 3 Ortho Group 4 Surgery	Group 5 Ortho Group 6 Surgery

MODULLE ONE: BSOF II
THEME ONE
TIME TABLE FOR WEEK THREE (March 28 – April 1/2022)

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
8:30-10:10 AM	TEST I (50) Anatomy (30) & Histology (20) Invigilators (Anatomy staff)	Embryology: First and second weeks of development Dr. Soressa	Embryology Third week of development Dr Soressa	Embryology: Fetal period Dr. Soressa	Embryology: Extra-embryonic Membranes and Placenta Dr. Soressa
10:10-10:30 AM		BREAK			
10:30:-12:10 AM		Embryology: 1 st & 2 nd wks of development Dr Soressa	Embryology: Embryonic Period Dr Soressa	Embryology: Fetal period Dr Soressa	Embryology: Multiple pregnancy Dr Soressa
12:10-1:30 PM		LUNCH			
1:30-4:30 PM		DR1: Group 3 DR2: Group 4 Mr. Abay & Mr Assaye	DR1: Group 5 DR2: Group 6 Mr. Abay & Mr Yossef	DR1: Group 1 DR2: Group 2 Dr. Girma & Mr Solomon A	DR1: Group 3 DR2: Group 4 Dr. Girma & Mr. Assaye
INSTRUCTOR		Hist. Lab 1: Group 5 1:30- 3:30 PM Hist. Lab 2: Group 6 3:30- 5:30 PM Mr. solomon	Hist. Lab 1: Group 1 1:30- 3:30PM Hist. Lab 2: Group 2 3:30- 5:30PM Mr. Fikre	Hist. Lab 1: Group 3 1:30- 3:30PM Hist. Lab 2: Group 4 3:30- 5:30PM Mr. Abay	Hist. Lab 1: Group 5 1:30- 3:30PM Hist. Lab 2: Group 6 3:30- 5:30PM Mr. Solomon
		Group 2 Ortho Group 1 Surgery	Group 4 Ortho Group 3 Surgery	Group 6 Ortho Group 5 Surgery	Group 1 Ortho Group 2 Surgery

MODULLE ONE: BSOF II
THEME TWO
TIME TABLE FOR WEEK FOUR (April 4–April 8 /2022)

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
8:30:-10:10 AM INSTRUCTOR	Embryology: Human Birth Defects I Human Birth Defects II Dr Soressa	Physiology: Excitable Tissue Mrs. Tigist	Physiology: Excitable Tissue Mrs. Tigist	Physiology: Autonomic Nervous System: Introduction; Physiologic anatomy; Mrs. Tigist	Physiology: Autonomic functions after spinal cord transaction Mrs. Tigist
10:10-10:30 AM	BREAK				
10:30:-12:10 AM INSTRUCTOR	Embryology: Development of Body Cavities, Mesenteries, and Diaphragm, Malformations Dr Soressa	Physiology: Excitable Tissue Mrs. Tigist	Physiology: Excitable Tissue Mrs. Tigist	Physiology: ANS Receptors of effector organs Activity pattern Mrs. Tigist	Physiology: Autonomic functions after spinal cord transaction Mrs. Tigist
12:10-1:30 PM	LUNCH				
1:30-4:30 PM	DR1: Group 5 DR2: Group 6 Mr. Solomon A & Mr. Assaye	Hist. Lab 1: Group 5 Hist. Lab 2: Group 6 Mr. Fikre	Hist. Lab 1: Group 3 Hist. Lab 2: Group 4 Dr. Soressa		
1:30-4:30 PM	Hist. Lab 1: Group 1	Group 2 Ortho	Group 5 Ortho	Group 4 Ortho	Group 6 Ortho
	Hist. Lab 2: Group 2 Mr. Abay	Group1 Surgery	Group 6 Surgery	Group 3 Surgery	Group 5 Surgery
	Group 3 Ortho	Group 3 Gyn-Obs	Group 1 Gyn-Obs	Group 5 Gyn-Obs	Group 4 Gyn-Obs
	Group 4 Surgery	Group 4 Pediatrics	Group 2 Pediatrics	Group 6 Pediatrics	Group 3 Pediatrics

MODULLE: BSOF II THEME TWO TIME TABLE FOR WEEK FIVE (April 11– April 15/2022)					
TIME	MONDAY	THUESD AY	WEDNES DAY	THURSDAY	FRIDAY
8:30:-10:10 AM Invigilators	TEST II (50) Embryology (28) and Physiology (22) Invigilators (Anatomy & Physiology staff)	Group 2 Gyn-Obs Group 1 Pediatrics	Group 3 Gyn-Obs Group 4 Pediatrics	Group 5 Gyn-Obs Group 6 Pediatrics	Group 2 Gyn-Obs Group 1 Pediatrics
12:10-1:30 PM		LUNCH			
1:30-4:30 PM		Group 6 Gyn-Obs Group 5 Pediatrics	Group 1 Gyn-Obs Group 2 Pediatrics	Group 4 Gyn-Obs Group 3 Pediatrics	Group 6 Gyn-Obs Group 5 Pediatrics

MODULE: BSOF II
THEME TWO
TIME TABLE FOR WEEK SIX (April 18– April 21/2022)

TIME	MONDAY	THUESDAY	WEDNESD AY	THURSDAY	FRIDAY
8:30:-10:10 AM	Oral exam	Oral exam	Study Period	FINAL EXAM (100) Anatomy (32) Histology (21) Embryology (26) Physiology (21) Anatomy & Physiology staff	Good Friday
Invigilators	DR1: Group-1*	DR1: Group-5*			
10:10-10:30 AM	Mr. Abay & Mr Solomon A	Dr Soressa & Mr Solomon A			
10:30:-12:10 AM	DR2: Group-2*	DR2: Group-6*			
	Mr. Assaye & Mr Yossef T	Mr Assaye & Mr Fikre B			
12:10-1:30 PM	Lunch				
1:30-4:30 PM	Oral exam	Oral exam	Study Period		
	DR1: Group-3*	DR1: Group-7*			
	Mr. Abay & Dr. Girma	Mr Solomon A & Mr Yossef T			
	DR2: Group-4*	DR2: Group-8*			
	Dr. Soressa & Mr. Yossef	Mr. Abay & Mr. Assaye			

N.B. * For oral exams, new grouping will be made and posted, oral exam will may exempt

VIII. Reading Resources

1. Moore KL, Dalley AF. Clinically Oriented Anatomy, 8th ed. Baltimore: Lippincott Williams & Wilkins, 2013
2. Henry Gray and Henry Vandyke Carter: Grey's Anatomy, 40th ed. Church. Livingston, London. 2008
3. Robert H. Whitaker and Neil R. Borley: Instant Anatomy, 4th ed. 2010
4. Frank H Netter: Atlas of Human Anatomy, 6th ed. Saunders, 2014.
5. Ross MH, Romrell, LJ & Kaye, G.I. Histology. A Text and Atlas. 6th ed. Williams & Wilkins. London
6. Junqueira, L.C., Carneiro, J. & Kelley, R.O. Basic Histology. 14th ed. Appleton & Lange. 2015
7. DiFiore's Atlas of Histology with Functional Correlations. Williams & Wilkins. London
8. Moore, K.L. The Developing Human: Clinically Oriented Embryology, 9th ed. W. B. Saunders, Philadelphia. 2011
9. Sadler, J.W. Langman's Medical Embryology, Williams & Wilkins, Baltimore.
10. Dissection Manual: Romanes, G.J., Cunningham's. with different volumes available in the Department of Anatomy
11. Textbook of Medical physiology, A. Guyton, 10th edition
12. Review of Medical physiology, William F. Ganong, 16th edition
13. Human Physiology, Seven Edition. Vander Sherman Luciano
14. Weaver, Robert F. *Molecular Biology*, 2nd ed. New York : McGraw-Hill, 2002.

IX. Lecture Topics

Theme	LECTURE TOPICS
	Anatomy
I	THEME ONE: Back (8 hrs) 1.1. General description; key features; components: skeletal, spinal nerves, surface anatomy; curvatures. 1.2. Typical and atypical vertebrae 1.3. Back ligaments and joints 1.4. Back musculature 1.5. Clinical correlation; Lumbar puncture; disc prolapse (to be taught together with clinicians)
I	THEME ONE: Upper Limb (16hrs) 2.1. General introduction to the upper limb. 2.2. The bones of the upper limb 2.3. Surface anatomy of the free upper limb – subcutaneous vessels and nerves 2.4. Pectoral region and breast 2.5. Axilla boundaries and contents

	2.6. Posterior compartment of the arm and forearm and dorsum of the hand 2.7. Anterior compartment of the arm and forearm; the cubital fossa 2.8. Palm of the hand 2.9. The joints of the pectoral girdle, elbow, wrist joints and other joints of the forearm and hand
I	THEME ONE: Histology (16 hrs) 3.1. Epithelial Tissue 1: Forms & Characteristics 3.2. Epithelial tissue 2: Covering & Glandular Epithelium 3.3. Connective tissue 1: Connective tissue proper 3.4. Connective tissue 2: Special Connective Tissue 3.5. Nervous Tissue 1: Neurons 3.6. Nervous tissue 2: Neuroglia 3.7. Peripheral Nervous System
I	THEME ONE: Embryology (20 hrs) 4.1. First Week of Development, Second Week of Development 4.2. Third Week of Development 4.3. Embryonic Period 4.4. Fetal Period 4.5. Extra-embryonic Membranes and Placenta 4.6. Multiple Pregnancy 4.7. Human Birth Defects I 4.8. Human Birth Defects I 4.9. Development of Body Cavities, Primitive Mesenteries, and the Diaphragm, Malformations
II	THEME TWO: Physiology (16 hrs)
	5. Cell excitation – 8 hours

	6. Autonomic Nervous System -4 hours 6.1. Introduction to ANS Physiologic anatomy of ANS Basic characteristics of the SANS and PANS Cholinergic and adrenergic fibers Mechanisms of transmitter secretion and removal Receptors of the effector organs Acetylcholine (muscarinic, nicotinic) Adrenergic receptors (alpha and beta) 6.2. The presynaptic control of transmitter release Effects on specific organs Function of the adrenal medulla Activity pattern of ANS Sympathetic and parasympathetic tone Effects of autonomic denervation Denervation supersensitivity Autonomic functions after spinal cord transaction
	7. Autonomic functions after spinal cord transaction – 4 hrs